
Spin and vacuum polarization remain invisible to charge conservation in lattice plasma dynamics

Anonymous Author

Abstract

Gauge symmetry can protect a plasma calculation while leaving its magnetic response physically incomplete. We separate these issues using an independently implemented variational scalar-electrodynamics lattice, verified against linear plasma spectra and laser-driven charge separation. Ring holonomy and a finite-amplitude electromagnetic condensate agree with analytic predictions under refinement, including a derived radial stability spectrum. Nevertheless, the converged scalar ground magnetic level differs from an aligned electron level by 34.1569%, and the tree-level vacuum has no polarization splitting. A static spin-sector action term and a low-frequency one-loop constitutive action restore the respective responses, with finest-grid errors below 0.0305% and 0.0024%, while preserving their charge and energy identities. Independently integrating the proper-time determinant reveals a 15.3342% bias in the simpler weak-field vacuum correction. The results distinguish conserved quantities, discretization error, and missing physical operators, and provide action-level repairs with explicit domains of validity.

1 Introduction

Exact charge conservation is a necessary numerical property of a plasma simulation, but it cannot establish that the simulation contains the correct magnetic response. Gauge symmetry constrains the continuity equation while leaving both the spin representation of matter and the quantum constitutive law of the vacuum unspecified. This distinction becomes consequential when collective plasma scales approach the Compton scale. The geometrical-optics route to kinetic theory then loses its usual scale separation [1], and laboratory pair plasmas [2, 3, 4, 5] motivate descriptions that can couple collective motion to relativistic quantum dynamics. Quantum kinetic approaches [6, 7, 8, 9, 10, 11, 12] and particle simulations with quantum processes [13, 14, 15] address different parts of this problem; interference between processes makes their domain of validity a physical question as well as a numerical one [16].

Real-time scalar electrodynamics offers a particularly direct setting in which to separate these questions. A link-based matter derivative and noncompact field-strength action give a local variational update with a discrete continuity identity [55, 36, 37]. This construction shares the geometrical motivation of lattice gauge theory [17] and geometric particle-in-cell methods [51, 52, 53, 54], while placing matter and fields on the same lattice. In a high-occupation, weak-coupling regime, classical-statistical evolution can approximate quantum dynamics [20, 21, 22, 23, 24]. The initial ensemble remains part of the physical model, not merely a source of numerical noise [25, 26, 27, 28]. The underlying nonequilibrium quantum formulation is instead expressed on a closed time contour [18, 19].

Scalar plasma theory already predicts dispersive collective effects beyond classical cold-fluid theory [33, 34, 38, 39, 40]. The later synthesis by Shi, Qin, and Fisch [63] discusses magnetized collective waves, including quantum corrections to cyclotron response and Faraday rotation. Those are distinct from the issue addressed here: whether invariants of a converged numerical trajectory can certify a magnetic matter representation or a vacuum constitutive law. Existing spinor-lattice calculations

[29, 30, 31] and the fermion-discretization literature [32] already make clear that choosing scalar matter avoids an important numerical difficulty by changing the physical theory. The magnetic spin term itself is standard [57, 58, 59], as is the scalar vacuum effective action [60]. We do not claim either as new physics.

We establish a quantitative separation between geometric fidelity and physical completeness within a common variational framework. A flat connection with nontrivial ring holonomy tests gauge equivalence without electromagnetic force. A self-consistent finite-amplitude circular condensate tests nonlinear coupling and conserved quantities, and yields an independently derived radial stability spectrum. Against these controls, magnetic matter levels and photon propagation in an unoccupied magnetized vacuum expose two different omitted responses. We derive and run local action modifications for the static spin sector and for the low-frequency one-loop vacuum, tracking which invariants survive. An additional distinction emerges between discretization convergence and convergence of the background-field expansion of the effective action.

The familiar plasma examples remain essential controls. We reproduce the unmagnetized spectra and the laser-driven transition from a subcritical wake to a supercritical charge-separating state studied in Ref. [55]. Wakefield and laser-acceleration mechanisms have a substantial classical literature [41, 42, 46, 47, 48], while particle production can involve mechanisms with different microscopic origins [43, 44, 45]. Wave-energy and spectral changes also respond to the evolving plasma [49, 50]. For this reason, our laser verification uses the published field and charge signatures without equating charge-sign reversal with a uniquely measured Schwinger rate. The many-particle character of the Klein–Gordon field requires precisely this care [35].

2 Methodology

2.1 A common gauge-invariant lattice action

We use natural units $m = c = \hbar = \epsilon_0 = 1$ and positive coupling $q = 0.3$. With the covariant spatial potential convention of Ref. [55], $E_l = \delta_t A_l - \delta_l A_0$ and $\mathbf{B} = -\text{curl}_h \mathbf{A}$. Vertices carry a complex scalar and edges carry a real, noncompact potential. A forward link derivative is

$$D_\mu^+ \phi_v = \frac{U_{v\mu} \phi_{v+\mu} - \phi_v}{h_\mu}, \quad U_{v\mu} = e^{-iq h_\mu A_{v\mu}}. \quad (1)$$

Our implementation extremizes precisely the Yee-field-strength action of Eqs. (14)–(15) of Ref. [55],

$$S_h = \sum_{n,v} \Delta t \Delta V \left[|D_0^+ \phi|^2 - \sum_l |D_l^+ \phi|^2 - m^2 |\phi|^2 + \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \right]. \quad (2)$$

We fix temporal gauge only after variation. Defining the link Laplacian $\Delta_A \phi_v = \sum_l (U_{vl} \phi_{v+l} - 2\phi_v + U_{v-l,l}^* \phi_{v-l})/h_l^2$, the explicit updates are

$$\phi^{n+1} = 2\phi^n - \phi^{n-1} + \Delta t^2 (\Delta_A \phi^n - m^2 \phi^n), \quad (3)$$

$$\mathbf{E}^{n+1/2} = \mathbf{E}^{n-1/2} + \Delta t (\text{curl}_h \mathbf{B}^n + \mathbf{J}^n), \quad \mathbf{A}^{n+1} = \mathbf{A}^n + \Delta t \mathbf{E}^{n+1/2}. \quad (4)$$

The covariant current and temporal-link charge are

$$\mathbf{J}_{vl}^n = \frac{2q}{h_l} \text{Im}(\phi_v^{n*} U_{vl}^n \phi_{v+l}^n), \quad \rho_v^{n+1/2} = \frac{2q}{\Delta t} \text{Im}(\phi_v^{n*} \phi_v^{n+1}). \quad (5)$$

Thus the positive-frequency initial scalar carries negative charge in this convention. An immobile background neutralizes it when required. We solve $\text{div}_h E^{1/2} = \rho^{1/2} + \rho_{\text{ion}}$ initially by a discrete cumulative sum, subtracting the mean electric field. The equations subsequently propagate this constraint without projection.

The same links give the identity

$$\delta_t \rho^n = \text{div}_h \mathbf{J}^n, \quad \delta_t \mathbf{B} = -\text{curl}_h \mathbf{E}, \quad \text{div}_h \mathbf{B} = 0. \quad (6)$$

Consequently global charge is conserved up to accumulated floating-point roundoff; magnetic identities follow from $d_h^2 = 0$. The divergence of B is identically zero in the reduced geometries.

We measure the nontrivial Faraday residual directly. Gauge covariance is tested both off shell, by transforming the complete spacetime action, and on shell, by evolving gauge-equivalent initial data. Independent finite differences of the action also check the scalar and gauge forces, avoiding a test that merely repeats the implementation.

One resolved coordinate is retained for the plasma demonstrations, with three potential components and periodic single-cell transverse directions. Those transverse links are not linearized: their contribution to Eq. (3) is $-4 \sum_{l=y,z} \sin^2(qhA_l/2)\phi/h^2$, and $J_l = -2q|\phi|^2 \sin(qhA_l)/h$. Every real-time update is local. No spectral solution or analytic frequency is substituted for evolution. The general free-lattice stability condition is $\Delta t^2[m^2 + 4 \sum_l h_l^{-2}] \leq 4$; the separate one-direction Courant conditions are not sufficient on a fully resolved multidimensional grid. Our time steps also resolve the field-induced matter frequency.

2.2 Energy, observables, and decision rules

For the coupled problem we record the bilinear energy used in the source paper, with its unspecified higher-order correction set to zero:

$$\mathcal{U}^{n+1/2} = \Delta V \sum_v \left[\left| \frac{\phi^{n+1} - \phi^n}{\Delta t} \right|^2 + m^2 \text{Re}(\phi^{n*} \phi^{n+1}) + \sum_l \text{Re}\{(D_l \phi^n)^* D_l \phi^{n+1}\} + \frac{1}{2} |\mathbf{E}^{n+1/2}|^2 + \frac{1}{2} \mathbf{B}^n \cdot \mathbf{B}^{n+1} \right]. \quad (7)$$

This is exactly conserved for the fixed quadratic operators used below, but has a finite discretization defect for nonlinear matter–field coupling. Variational does not imply exact energy conservation at fixed time step. Charge drift means $\max_t |Q(t) - Q(t_0)|/|Q(t_0)|$; energy drift is defined analogously. For the neutral vacuum the charge ratio is undefined, so we report the absolute drift instead.

All decisions below were recorded in `logs/protocol.json` before production. Frequency fits use an FFT to locate a peak, followed by least-squares fitting of a sinusoid with a constant offset. A reference frequency is never used to drive the simulations. The plasma spectra use a Hann window and an ensemble average of power, with independently detected local maxima. Their acceptance width is one Fourier bin, a resolution limit rather than a fitted tolerance. For the resolved single-mode probes, the relative-frequency tolerance is 0.5%. It is deliberately larger than the leading Taylor errors of the link Laplacian and temporal difference on the finest meshes; refinement must also exhibit the expected quadratic trend. The gauge and charge tolerances are 10^{-9} , allowing accumulated roundoff while remaining far below any physical signal. The nonlinear-orbit energy tolerance is 10^{-3} . Magnetic model failure means a ground-level error exceeding 10%, a shift large compared with discretization errors. Vacuum model failure means losing more than half the predicted polarization splitting; its repair must recover that splitting within 1%. We distinguish numerical agreement with the weak-field action from agreement with the full zero-frequency one-loop action. The latter comparison uses the same tolerance and an additional protocol recorded before its production runs.

2.3 Holonomy without electromagnetic force

On a ring of circumference L with flat connection $A_x = \theta/(qL)$, separation of variables gives

$$E_\ell(\theta) = \sqrt{m^2 + [(2\pi\ell - \theta)/L]^2}, \quad E_{\ell+1}(\theta + 2\pi) = E_\ell(\theta). \quad (8)$$

This is a gauge-holonomy effect [56], not an unmagnetized plasma wave. The connection is prescribed: an external source holds the ring flux fixed. A static source exchanges no energy with the scalar here. We evolve a plane wave with a generic initial phase velocity m , so that both signs of the lattice frequency can be present, and compare its oscillation frequency with Eq. (8). We then apply a smooth, large-amplitude periodic gauge transformation together with one unit of winding, transform the links by the exact discrete difference, and evolve again. The *primary diagnostic is gauge invariance*: the transformed trajectory must equal $e^{iq\alpha}\phi$ to the stated tolerance, in addition to matching the physical frequency. A large bare link argument does not by itself constitute a physical failure.

2.4 A finite-amplitude electromagnetic condensate

A spatially uniform, neutralized scalar admits the continuum orbit

$$\phi = R e^{-i\omega t + i\beta}, \quad (A_y, A_z) = a(\cos(\Omega t + \beta), \sin(\Omega t + \beta)), \quad \omega^2 = m^2 + q^2 a^2, \quad \Omega^2 = 2q^2 R^2. \quad (9)$$

Substitution into both field equations proves the prediction: the scalar sees a constant $A_y^2 + A_z^2$, while its constant modulus supplies the electromagnetic restoring current. Both fields are dynamical, including their backreaction. We test a family extending to $qa/m = 2$, where the matter frequency is substantially displaced from its rest frequency. The *primary diagnostic is charge and energy conservation* under strong coupling. We additionally require both independently measured frequencies to meet the frequency tolerance. The transverse Wilson links break continuous rotation symmetry at finite spacing, making this an actual convergence test rather than an exact discrete solution.

There is also a nontrivial local stability prediction. Holding the two angular momenta fixed and perturbing the scalar and gauge radii gives

$$\frac{d^2}{dt^2} \begin{pmatrix} \delta R \\ \delta a \end{pmatrix} = - \begin{pmatrix} 4\omega^2 & 2q^2 Ra \\ 4q^2 Ra & 4\Omega^2 \end{pmatrix} \begin{pmatrix} \delta R \\ \delta a \end{pmatrix}. \quad (10)$$

Both eigenvalues are positive. Their square roots predict two radial sidebands, which we check with a small displacement and a longer record. This supplementary prediction distinguishes a stable coupled orbit from two unrelated imposed sinusoids.

2.5 Magnetic levels and the missing spin coupling

Take a prescribed uniform magnetic field, $qB > 0$, and a Fourier sector $p_y = p_z = 0$. The continuum scalar operator is $-\partial_x^2 + (qBx)^2 + m^2$, so its levels are

$$E_{n,\text{sc}}^2 = m^2 + (2n+1)qB, \quad E_{n,s}^2 = m^2 + (2n+1)qB - sqB. \quad (11)$$

The second expression is the spin-resolved Dirac prediction, obtained by squaring the Dirac operator; $s = +1$ is the aligned sector [57, 58, 59]. The scalar ground state therefore cannot describe an aligned electron ground state even with an arbitrarily fine mesh. This is a boundary of applying a spinless plasma model to electrons, not an error of scalar QED as a theory of bosons. Its *primary diagnostic is the matter–gauge coupling*: orbital minimal coupling alone cannot create the Pauli term.

In this static, spin-diagonal sector a local action-level repair is

$$S_{h,s} = S_h + \sum_{n,v} \Delta t \Delta V \, sqB_{z,v} |\phi_{s,v}|^2, \quad \delta_t^2 \phi_s = \Delta_A \phi_s - (m^2 - sqB_z) \phi_s. \quad (12)$$

The field strength is averaged to the vertex if needed; it is constant in our experiment. The temporal-link charge is unchanged. If the magnetic field is varied, this action also generates the magnetization $M_z = sq|\phi_s|^2$: Maxwell–Ampère becomes $\partial_t \mathbf{E} = \nabla \times (\mathbf{B} - \mathbf{M}) + \mathbf{J}_{\text{orb}}$ in the covariant-current convention. This curl is divergence-free and cannot violate charge conservation. We implement the fixed-field restriction of Eq. (12); no self-consistent magnetization run is claimed. The energy includes $-sqB|\phi_s|^2$ and is time invariant for a static background. The lattice retains both exact global charge and its quadratic bilinear energy. The Bianchi construction is unchanged. This repair restores the magnetic spin spectrum, but does not implement fermion occupation constraints or the electric spin-mixing terms of a general Dirac theory.

On the actual strip mesh the orbital potential is $4 \sin^2(qBxh/2)/h^2$, directly inherited from the transverse Wilson link, not the continuum harmonic potential. Its expansion gives a useful independent error estimate:

$$\delta E_n^2 = -\frac{(qB)^2 h^2}{8} (2n^2 + 2n + 1) + O(h^4). \quad (13)$$

The two equal contributions come from the longitudinal difference and the transverse link; adding the constant spin term changes neither. Temporal dispersion adds $E^3 \Delta t^2 / 24$ to the measured frequency. This predicts both the order and sign of refinement errors.

2.6 Virtual polarization of an empty magnetized vacuum

At tree level, the unoccupied mean-field state $\phi = 0$ leaves an exactly linear Maxwell action, independently of the static magnetic background. It therefore predicts identical photon speeds for the two transverse polarizations. The renormalized quantum vacuum has a nonzero virtual polarization even when the mean scalar vanishes. In the low-energy, weak-field scalar theory, its leading effective Lagrangian is

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{scalar QED}} + C [7(\mathbf{E}^2 - \mathbf{B}^2)^2 + 4(\mathbf{E} \cdot \mathbf{B})^2], \quad C = \frac{q^4}{5760\pi^2 m^4}. \quad (14)$$

The coefficient follows from the scalar one-loop determinant [60]; it differs from the electron coefficient. The *primary diagnostic is dispersion fidelity*. Refining a tree-level Maxwell update cannot generate a term absent from its action. This statement concerns the tree-level closure used here, not all classical-statistical methods: vacuum ensembles with an appropriate regulator and renormalization can retain quantum information [25, 26, 28, 62].

For $\mathbf{B}_0 = B\hat{z}$ and propagation along x , expand Eq. (14) to second order in the probe. The fully specified discrete repair that we implement is

$$S_{h,\text{probe}} = \frac{1}{2} \sum_{n,i} \Delta t h \sum_{a=\perp,\parallel} [\epsilon_a (\delta_t A_a)^2 - \nu_a (\delta_x A_a)^2], \quad \epsilon_a \delta_t^2 A_a = \nu_a \delta_x^2 A_a, \quad (15)$$

where $a = \perp$ means E_y and $a = \parallel$ means E_z . Differentiating the Lagrangian, rather than fitting material constants, gives

$$\begin{aligned} (\epsilon_{\perp}, \nu_{\perp}) &= (1 - 28CB^2, 1 - 84CB^2), & v_{\perp}^2 &= \frac{1 - 84CB^2}{1 - 28CB^2}, \\ (\epsilon_{\parallel}, \nu_{\parallel}) &= (1 - 20CB^2, 1 - 28CB^2), & v_{\parallel}^2 &= \frac{1 - 28CB^2}{1 - 20CB^2}. \end{aligned} \quad (16)$$

We measure $(\omega_{\parallel} - \omega_{\perp})/k$ directly from two evolved waves and compare with $v_{\parallel} - v_{\perp}$. No grid-dispersion correction is subtracted from the reported splitting. Its leading relative error is $-k^2 h^2/24 + O(\Delta t^2, h^4)$, which justifies the tolerance and predicts the convergence slope.

To avoid confusing a resolved F^4 action with a converged quantum constitutive law, we also derive coefficients without truncating the background-field strength. Write the renormalized scalar determinant as

$$\mathcal{L}_1 = \frac{1}{16\pi^2} \int_0^\infty \frac{ds}{s^3} e^{-m^2 s} \left[\frac{(qas)(qbs)}{\sin(qas) \sinh(qbs)} - 1 + \frac{q^2 s^2}{6} (b^2 - a^2) \right], \quad (17)$$

where $a^2 - b^2 = E^2 - B^2$ and $ab = \mathbf{E} \cdot \mathbf{B}$. Let $\ell(B) = \mathcal{L}_1(0, B)$ and $f(z) = z/\sinh z$. Its second variation gives the same probe action, now with

$$\begin{aligned} \epsilon_{\perp} = \nu_{\parallel} &= 1 - \ell'(B)/B, & \nu_{\perp} &= 1 - \ell''(B), \\ \epsilon_{\parallel} &= 1 + \frac{q^2}{48\pi^2} \int_0^\infty \frac{ds}{s} e^{-m^2 s} [f(qBs) - 1]. \end{aligned} \quad (18)$$

The prediction is again $\Delta v = \sqrt{\nu_{\parallel}/\epsilon_{\parallel}} - \sqrt{\nu_{\perp}/\epsilon_{\perp}}$. We evaluate these integrals with 45-digit arithmetic before running the corresponding lattice waves. Equations (17)–(18), not fitted trajectories, specify the final vacuum repair. The F^4 calculation is retained as a separate approximation whose accuracy can be assessed. Both neglect finite-frequency derivative corrections and higher loops.

The general effective Maxwell equation is $\partial_t \mathbf{D} = \nabla \times \mathbf{H} + \mathbf{J}$, with $\mathbf{D} = \partial \mathcal{L}_{\text{eff}}/\partial \mathbf{E}$ and $\mathbf{H} = -\partial \mathcal{L}_{\text{eff}}/\partial \mathbf{B}$; Gauss law uses $\mathbf{D}$. It preserves gauge symmetry, charge continuity, and the Bianchi identities. Our quadratic probe action additionally conserves $\frac{1}{2} \sum_i h [\epsilon_a E_a^2 + \nu_a b_a^n b_a^{n+1}]$ exactly up to roundoff. The modified constitutive energy replaces the Maxwell energy; the latter alone is not the invariant of the repair. External static backgrounds remove momentum exchange with their sources from the simulated system. Neither magnetic repair claims conservation of that omitted total momentum.

3 Experimental Setup

The implementation is written independently in Python and Numba, using double-precision evolution throughout. The action audit, predictions, configurations, seeds, timestamps, complete conservation traces, and postprocessing scripts accompany the paper. Full-field archives of both plasma demonstrations retain every time slice of ϕ and all three A components in single precision; the diagnostics and restart states remain double precision. These archives are not the source of the roundoff-level conservation claims. The analytic probes retain full observable histories and either full fields or a stated field stride.

The linear-wave demonstration uses the source-paper density $\omega_p = 0.85$, box length 20.48, independent potential noise of standard deviation 10^{-4} , and a cold scalar with random global phase. There are 16 realizations per spacing, with $h = 0.08, 0.04, 0.02$ and $\Delta t = h/2$. The original observation time 20.48 is retained for direct comparison. An additional 81.92 record at $h = 0.04$ tests a weak longitudinal line that is difficult to separate in the shorter window. The transverse prediction is $\omega^2 = \omega_p^2 + k^2$; the two nonzero longitudinal branches solve

$$z^2 - (2k^2 + 4m^2 + \omega_p^2)z + k^4 + \omega_p^2 k^2 + 4m^2 \omega_p^2 = 0, \quad z = \omega^2. \quad (19)$$

The immobile-ion case has no independently propagating ion-acoustic branch. A zero-frequency component is not counted as a third dispersive mode.

The laser demonstration uses a periodic box of length 600, a density plateau of width 100 centered at 340, Gaussian ramps of width 20, and peak density 1. We initialize the scalar as in Ref. [55], then phase-mix for a seed-dependent time uniformly chosen between 160 and 185. A right-moving Gaussian pulse is injected with center 130, width 20, carrier frequency 0.7, and $q \max |A_y| = a = 1$ or 16. The evolution lasts 360, before the principal pulse wraps around the box. We use three realizations at each of $h = 0.16, 0.08, 0.04$, with $\Delta t = h/8$. These ensembles are smaller than the source paper's ensembles; means and sample standard deviations are reported rather than assuming identical precision. The prescribed reproduction signature is a subcritical wake for $a = 1$ and a supercritical longitudinal field for $a = 16$, together with positive scalar charge exceeding 1% of the initial charge magnitude. Positive charge is the source paper's antiparticle-like diagnostic, not an asymptotic particle-number measurement or an identification of a unique production mechanism.

The ring has $L = 20$, $\ell = 2$, $\theta = 1.1$, and $h = 0.2, 0.1, 0.05$, with $\Delta t = h/5$ and duration 80. Its large gauge function is $q\alpha = 100 \sin(2\pi x/L) + 0.7 \cos(6\pi x/L)$, followed by the winding transformation. The nonlinear condensate uses $R = 2$, $qa = 0.3, 1, 2$, $h = 0.16, 0.08, 0.04$, $\Delta t = h/8$, and duration 80. The additional radial perturbation is $\delta R/R = 0.01$ at $qa = 1$ and is observed for 240. Its initial scalar angular momentum is held fixed.

The magnetic strip has length 20, homogeneous $qB = 0.8$, and homogeneous Dirichlet scalar boundary values. We evolve Gaussian–Hermite profiles for levels $n = 0, 1, 2$ with a generic initial phase velocity m , using $h = 0.2, 0.1, 0.05$, $\Delta t = h/10$, and duration 100. The exponentially small wavefunction at the boundaries makes this a bulk Landau-level measurement. The vacuum probe has $qB = 0.3$, wave number $k = 0.05$, ring length $2\pi/k$, amplitude 10^{-4} , and $N = 64, 128, 256$ sites. Its time step is $h/20$ and duration is 800. Coarse vacuum spacings need only resolve the photon wavelength: no massive field is evolved in this effective probe action. All four analytic studies use phase seeds 11, 23, 37. Their phase variation checks reproducibility and gauge or rotation dependence; its standard deviation is not an estimate of quantum uncertainty. The magnetic-strip archives express charge in units of q ; reported relative drifts are independent of that normalization.

4 Experimental Results

Table 1 reports the reference linear mesh and the finest meshes of the other studies, with the worst drift over the relevant realizations. Every entry is generated from the archived trajectories or their analysis logs. Figure 6 separates frequency errors that decrease under refinement from missing physical responses that remain finite. Ensemble statistics at all resolutions appear in Table 2.

Table 1: Physical predictions, decision criteria, and measured outcomes. All tolerances are specified in Sec. 2. Categories distinguish implementation verification, scope within the scalar model, and boundaries of a physical approximation. For repaired cases the last two columns describe the repair. The vacuum charge entry is an absolute zero because a relative error for $Q_0 = 0$ is undefined.

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$\ \Delta Q/Q\ $	$\ \Delta \mathcal{U}/\mathcal{U}\ $
Linear plasma	verification	Unmagnetized waves	$\omega_T^2 = \omega_p^2 + k^2$; Eq. (19)	One Fourier bin	Both branches resolved; maximum 0.4994 bin error	1.05×10^{-14}	8.26×10^{-14}
Laser slab	verification	$a = 1, 16$	Subcritical wake; supercritical charge separation	$ E_x /E_c < 1$, then > 1 ; $Q_+/ Q_0 > 0.01$	Peak fields 0.2109, 1.9655; intense-pulse positive fraction 7.8266%	4.87×10^{-14}	3.55×10^{-4}
Ring flux	scope	Nonzero holonomy, $F = 0$	Eq. (8)	0.5% frequency; 10^{-9} gauge	0.0003% frequency error; gauge residual 2.75×10^{-13}	2.03×10^{-14}	1.91×10^{-14}
Circular condensate	scope	Nonlinear dynamical fields	Eq. (9)	0.5% frequency; 10^{-9} charge; 10^{-3} energy	Worst frequency error 0.0421% across amplitudes	6.70×10^{-13}	5.06×10^{-7}
Magnetic matter	boundary	Aligned electron levels	Eq. (11)	Failure $> 10\%$; repair $< 0.5\%$	Scalar ground error 34.1569%; spin repair error 0.0305% over levels	2.18×10^{-13}	2.18×10^{-13}
Magnetic vacuum	boundary	Polarized low-energy photons	Eq. (18)	Failure $> 50\%$; repair $< 1\%$ of splitting	Tree level loses all splitting; full-loop repair error 0.0024%	0 (abs.)	6.43×10^{-13}

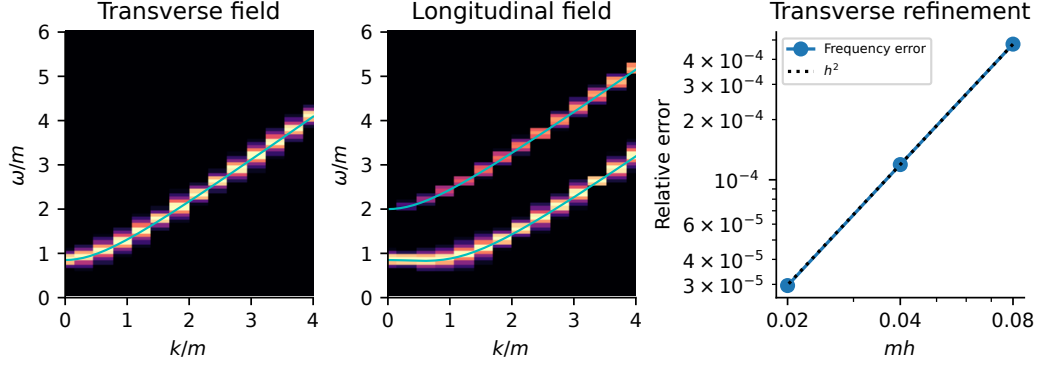


Figure 1: Left and center: ensemble-averaged electric-field spectra from the longer linear record at $h = 0.04$. Color is log power normalized separately at each wave number, ranging from -5 to 0 ; cyan curves are continuum predictions. Both spectra come from the actual potential histories, not from synthetic spectral lines. The upper longitudinal branch is weaker and requires a longer window at small wave number. Right: refinement of the fitted transverse frequency at Fourier index 6, using the original observation duration. The dotted curve is a quadratic reference slope.

4.1 Plasma spectra and laser-driven charge separation

The transverse and both nonzero longitudinal spectral ridges follow their analytic predictions (Fig. 1). The largest resolved deviation in the longer record is 0.4994 Fourier bins. At the shortest nonzero wave number, the short record does not separate the weak upper branch from leakage of the dominant lower branch. The longer record resolves it without modifying the initial noise or restricting the peak search to a prediction-centered interval. This distinguishes spectral resolution from a missing dynamical branch. The action-gradient checks agree to relative errors 7.61×10^{-10} for the scalar and 3.05×10^{-10} for the gauge force; the off-shell gauge transformation leaves the action unchanged at the recorded precision.

The finest laser ensemble gives maximum longitudinal fields $0.2109E_c$ and $1.9655E_c$ at moderate and intense amplitudes, respectively, with sample standard deviations $1.22 \times 10^{-4}E_c$ and $3.82 \times 10^{-5}E_c$. The maximum integrated positive-charge fractions are $(0.0218 \pm 3.98 \times 10^{-5})\%$ and $(7.8266 \pm 0.0979)\%$. The prescribed transition is therefore reproduced. Figure 2 shows the expulsion, wake, and charge-sign structures directly. The field and positive-charge criteria remain satisfied across the refinement series (Fig. 3). This verifies the qualitative demonstrations; it does not reproduce the source paper's precise ensemble-averaged peak fields, scattering amplitudes, or phase-mixing distribution, for which the initialization record was not distributed.

The electromagnetic energy losses are $(6.6994 \pm 0.0168)\%$ and $(8.2683 \pm 0.0101)\%$. The intense pulse also broadens its spectrum (Fig. 4); for the displayed seed the root-mean-square wave-number width changes from 0.0353 to 0.1354. These are emergent observables, not acceptance targets. A charge-density sign is not an on-shell particle counter, so we make no quantitative pair-yield or microscopic-channel claim.

4.2 Topology and a nonlinear orbit within scalar electrodynamics

The ring frequency is 1.1527 with a phase-ensemble standard deviation 6.84×10^{-11} , and its worst finest-mesh error is 0.0003%. The largest trajectory mismatch after the exact gauge transformation is 2.75×10^{-13} . The test includes a maximum bare link argument of 6.3653, exceeding 2π , without changing the physical trajectory. What matters here is the covariant phase difference, not the size of a gauge-dependent potential.

The nonlinear condensate satisfies both frequency relations through the whole amplitude family, with a worst finest-grid error of 0.0421%. Its charge and energy drifts remain below their specified bounds. Figure 5 overlays the evolved fields with the continuum solutions. The displaced orbit has fitted radial frequencies 1.5388 and 2.9174, agreeing with the eigenvalues of Eq. (10) to 0.0126% on the

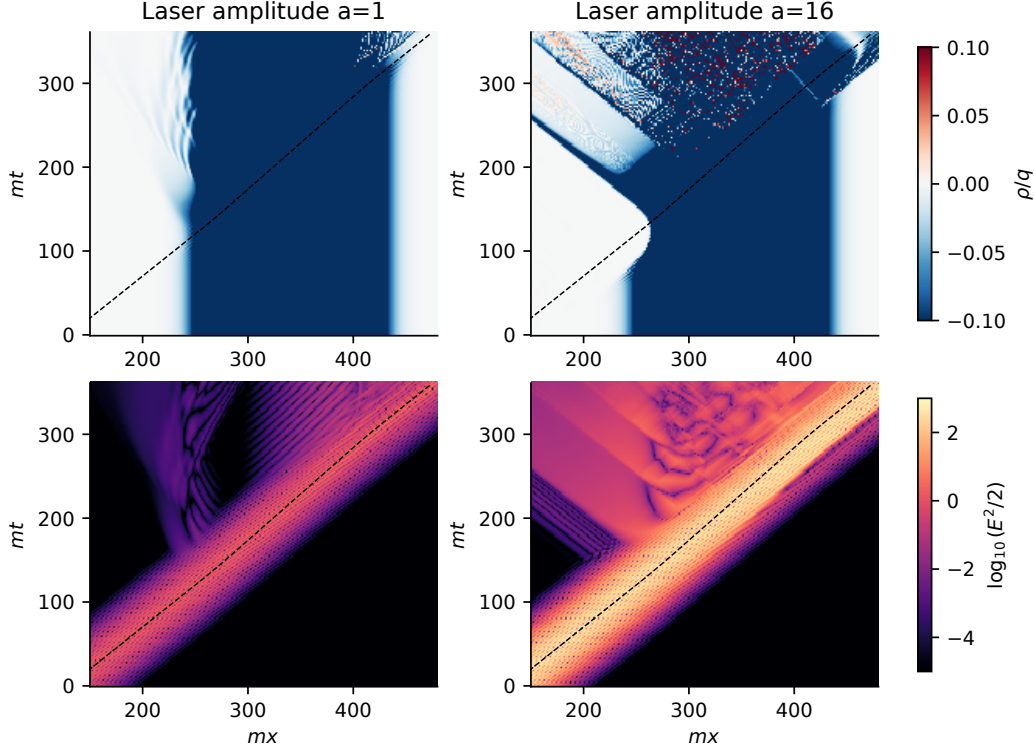


Figure 2: One finest-grid realization of each laser regime. Top: scalar charge density divided by q , with a common diverging scale saturated at ± 0.1 (red positive, blue negative). Bottom: $\log_{10}$ of electric field energy density, on a common scale from -5 to 3 . Dashed curves are geometric-optics trajectories through the initial static density; their departure from strong-field dynamics is not used as an accuracy criterion. Full field histories are retained.

finest grid. This agreement supports the coupled restoring mechanism and its stability, rather than only the unperturbed kinematics.

4.3 Responses that charge conservation cannot supply

The unmodified scalar magnetic ground level differs from the aligned electron target by 34.1569% on the finest grid. Its continuum limit remains $\sqrt{m^2 + qB}$ rather than m . The repaired ground level is 0.9999, with phase-ensemble standard deviation 9.26×10^{-16} ; the maximum error over the three tested levels is 0.0305%. The sign and quadratic scaling of the residual errors agree with Eq. (13). Adding the missing spin coupling preserves the charge and fixed-background energy identities.

The tree-level vacuum has identically zero polarization splitting at every resolution. The F^4 repair produces a nonzero splitting and converges accurately to its own prediction, 3.4196120×10^{-6} . However, the independent proper-time determinant predicts 2.9649603×10^{-6} , exposing a 15.3342% bias in the weak-field truncation at the chosen background. The full one-loop constitutive repair gives $2.9648877 \times 10^{-6} \pm 4.26 \times 10^{-15}$, with a worst relative error 0.0024%. Figure 6 displays all three outcomes: mesh refinement removes the numerical dispersion error but removes neither the absent loop nor the background-field truncation bias.

5 Analysis

The computations expose three logically different limits. A lattice can converge to its action, an approximate action can converge to an effective field theory, and that field theory can approximate the intended physical system. Charge conservation constrains the first limit. The magnetic spin

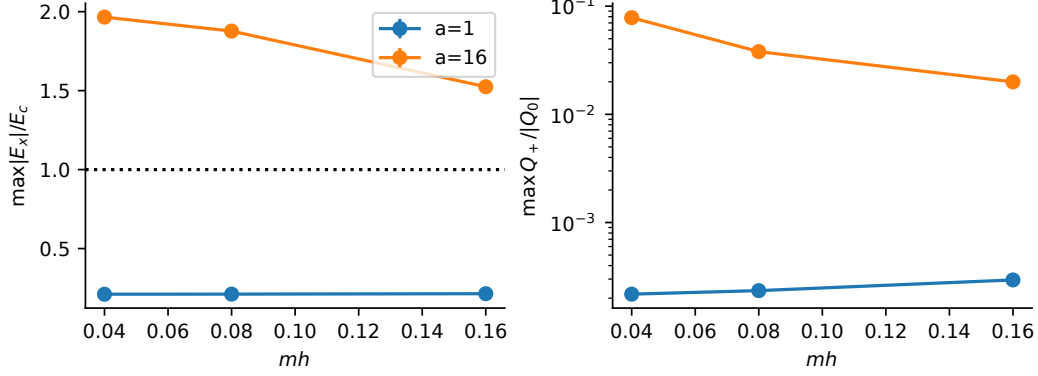


Figure 3: Laser refinement study. Markers and bars show the mean and sample standard deviation of the three realizations. Left: maximum longitudinal field; the dotted line is E_c . Right: largest integrated positive-charge fraction. Nonlinear amplitudes remain mesh-sensitive even when charge conservation is excellent. The two qualitative reproduction decisions remain unchanged.

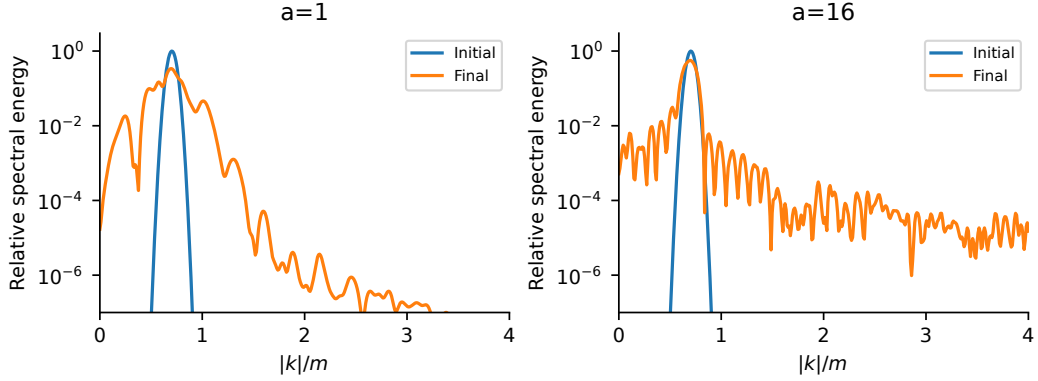


Figure 4: Initial and final transverse electromagnetic spatial spectra from the displayed laser realizations, normalized by each initial maximum. Both propagation directions contribute to $|k|$. The intense pulse develops a broader distribution; no direction-resolved scattering rate is inferred from these spectra.

discrepancy belongs to the choice of matter representation, while the missing vacuum splitting belongs to the quantum closure. The intermediate F^4 result supplies a particularly sharp counterexample to using small invariant drift as a surrogate for any of the other limits: its quadratic energy is conserved as accurately as that of the full one-loop repair, although their continuum predictions differ visibly.

This distinction also clarifies what a resolution condition should mean. For a smooth charged configuration, the link energy depends on $\arg \phi_{i+1} - \arg \phi_i - qhA_i$. A condition on qhA_i alone is not gauge invariant. A flat, large pure-gauge contribution can make that quantity arbitrarily large without changing either covariant matter gradients or electromagnetic fields, as the ring trajectories demonstrate. Moreover, the noncompact Maxwell part of Eq. (2) is not invariant under arbitrary, independent integer shifts of every edge potential: such shifts generally change the plaquette curvature. Periodicity of the matter link therefore does not make the entire noncompact gauge action a collection of independent angular variables. This observation does not remove ultraviolet constraints; it identifies the covariant momenta and curvature, rather than a gauge-dependent potential magnitude, as the physically relevant resolved quantities.

The nonlinear orbit provides a complementary lesson. Its two continuum rotations are sustained by mutual coupling, yet the finite transverse links retain only the lattice rotation symmetry. Small modulation of the orbit is consequently expected even without a physical instability. Equation (10) predicts stable coupled radial modes, and the refinement study distinguishes those physical frequencies

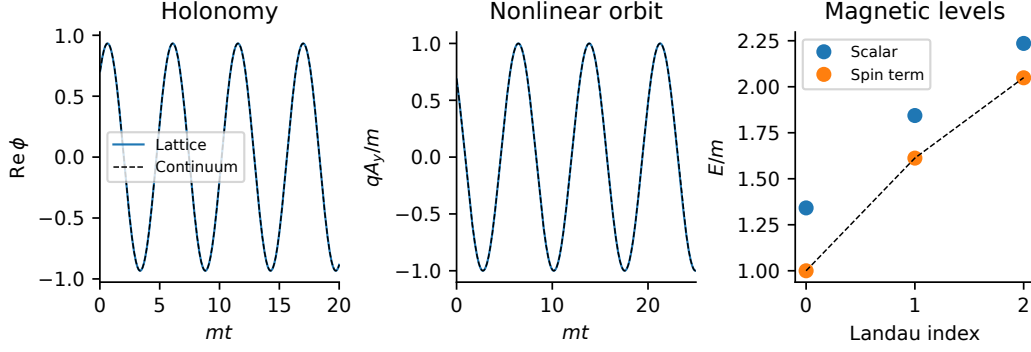


Figure 5: Independent analytic comparisons. Left: ring trajectory and continuum solution. Center: a transverse component of the self-consistent nonlinear orbit. Right: the evolved lowest magnetic frequencies before and after adding the spin term; the dashed curve is the aligned electron prediction. Markers and curves coincide where agreement is accurate.

Table 2: Phase-ensemble statistics at every resolution. Frequencies and energies are in units of m ; the vacuum splitting is displayed in units of 10^{-6} . The orbit row uses its electromagnetic frequency at the largest amplitude. Standard deviations reflect the three phase seeds, not quantum sampling uncertainty.

Regime	Spacing / size	Observable mean	Sample std.	Max. error
Ring frequency	0.2	1.1526359	5.32e-11	0.00469%
Ring frequency	0.1	1.1526764	6.32e-11	0.00117%
Ring frequency	0.05	1.1526866	6.84e-11	0.00029%
Orbit frequency ($qa = 2$)	0.16	0.8429929	1.88e-04	0.67753%
Orbit frequency ($qa = 2$)	0.08	0.8471497	4.48e-05	0.16846%
Orbit frequency ($qa = 2$)	0.04	0.8481838	1.10e-05	0.04206%
Spin ground energy	0.2	0.9984174	6.13e-16	0.15826%
Spin ground energy	0.1	0.9996042	7.81e-16	0.03958%
Spin ground energy	0.05	0.9999010	9.26e-16	0.00990%
Vacuum Δv (10^{-6})	64	2.9637798	2.57e-09	0.03981%
Vacuum Δv (10^{-6})	128	2.9646661	2.83e-09	0.00992%
Vacuum Δv (10^{-6})	256	2.9648877	4.26e-09	0.00245%

from discretization-induced modulation. The error formula for the Landau levels makes a similar separation explicit: the orbital stencil error is unchanged by the constant spin term, whereas the physical ground-level shift remains finite as the stencil error vanishes. A useful diagnostic is therefore to ask which part of a discrepancy has an allowed scaling with spacing, and which part requires a different operator.

The action-level repairs explain why physical improvement need not cost geometric structure. The magnetic term depends on B and a neutral bilinear, so its additional current is a magnetization curl. The vacuum term depends only on field strength, so it changes constitutive coefficients without changing the gauge symmetry. Both preserve the same continuity mechanism. What changes is the energy functional and, for prescribed backgrounds, the set of degrees of freedom included in the energy and momentum budget. Monitoring the original Maxwell energy after changing the constitutive action would test the wrong invariant.

There are material limits to these conclusions. The static spin-sector repair does not constitute a general spinor plasma solver: it omits electric spin mixing, Pauli blocking, and many-body fermion statistics. The vacuum repair is the low-frequency second variation of a scalar one-loop determinant, not an electron vacuum response or a dynamical pair-production model. Combining it with spinor matter would require the corresponding spinor determinant and a consistent separation of explicit particles from integrated-out fluctuations. Finite-frequency polarization can be nonlocal [61]; higher-loop effects are also outside the calculations. These restrictions are physical approximations, not

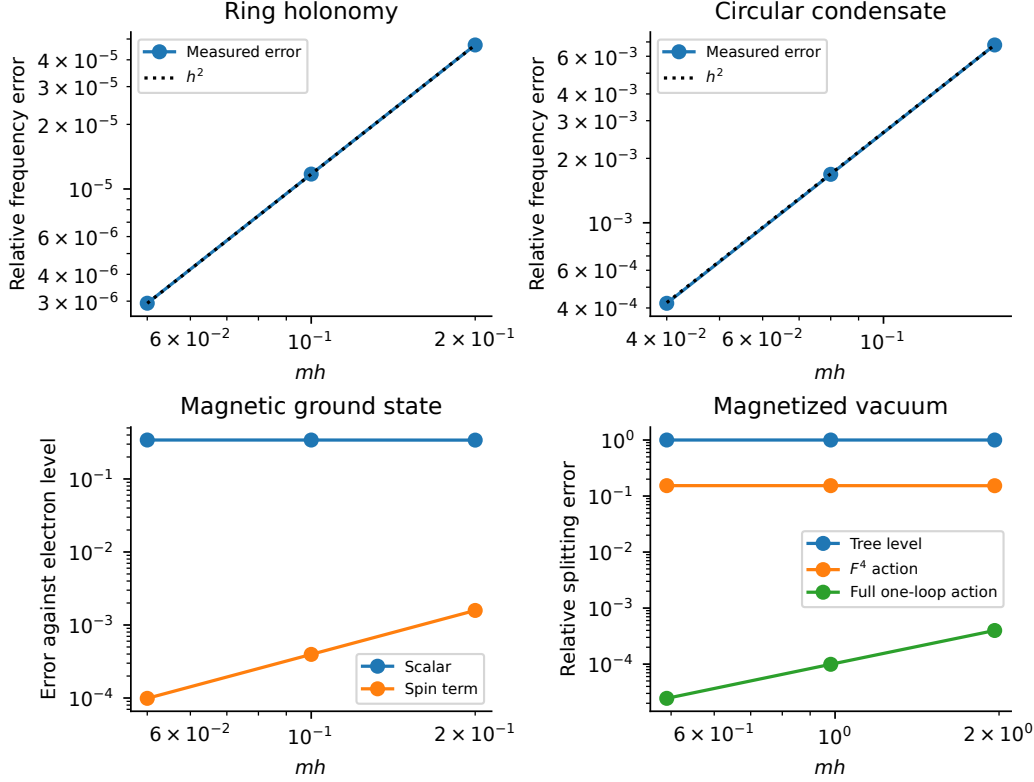


Figure 6: Convergence separates discretization error from missing response. Top: scalar-model frequency errors, with quadratic reference slopes. Bottom left: error against the aligned electron ground state. Bottom right: splitting error against the full zero-frequency scalar one-loop determinant. The intermediate F^4 action converges to a biased physical value despite excellent numerical accuracy.

effects that a smaller time step can cure. Classical-statistical vacuum ensembles provide another route, but their ultraviolet treatment requires care [26, 62].

The analytic probes have tightly controlled phase ensembles, whereas the laser reproduction has only three phase-mixing samples per mesh and substantial nonlinear mesh dependence in its amplitudes. The reproduction establishes the stated transition, not a precision pair-yield benchmark. No continuum-extrapolated laser production rate, experimental prediction, or claim about spinor pair creation follows. The calculations are spatial reductions; they do not establish behavior in a turbulent multidimensional plasma. Quantum digitizations of scalar gauge theories [64] could address different aspects of the quantum state, but would still need physically appropriate operators and invariant-aware observables.

6 Conclusion

Gauge-invariant variational evolution accurately transports a charged field through nontrivial holonomy and sustains a strongly coupled electromagnetic condensate. It also conserves charge while omitting magnetic spin response and virtual vacuum polarization. These omissions survive lattice refinement and are repaired, in the stated static-spin and low-frequency sectors, by adding the missing terms at the action level. The vacuum calculation further shows that an effective-action truncation can produce a converged but biased response. Conservation, discretization convergence, and physical completeness must therefore be assessed independently. The accompanying implementations and complete run records make those distinctions directly reproducible.

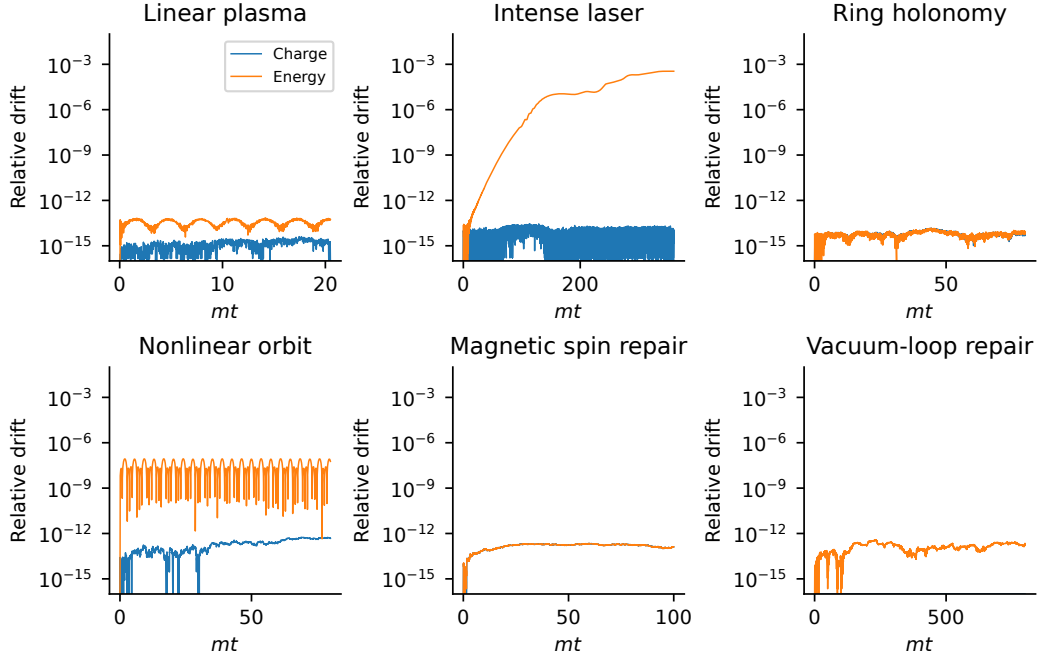


Figure 7: Charge and energy histories for one realization per regime. The magnetic panels show the final repairs. The vacuum’s identically zero charge is drawn at the plotting floor; this is an absolute zero, not a meaningful relative-charge measurement. A common vertical scale makes the finite nonlinear energy defect and the roundoff-level quadratic invariants distinguishable.

Data and code availability. All reported trajectories, configurations, phase seeds, diagnostics, derived spectra, figure scripts, and machine-readable numerical claims are supplied in `logs/` and `code/`. The execution history and rejected approaches are documented in `attempts_log.md`.

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